Supplementary Material

Anonymous CVPR submission

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Blade	Airfoil span (mm)	Cases	Visible dents	AUROC	Found (%)	False alarm (%)
1	24	76	31	0.98	100	100
2	26	76	31	0.72	100	100
3	31	76	31	0.64	97	100
4	40	76	31	0.66	94	82
5	52	76	31	0.93	84	6
6	70	76	31	0.86	87	15

Table 1. Ours on each CAD6 blade, smallest to largest airfoil (frozen threshold).

Variant (CAD6, 48 cases)	AUROC	Found (%)	False alarm (%)
Full method	0.71 [0.62, 0.83]	92 [83, 100]	78 [56, 100]
No camera correction	0.90 [0.81, 0.98]	92 [83, 100]	56 [33, 67]
No stain model	0.73 [0.64, 0.84]	96 [88, 100]	89 [78, 100]
Random starts	0.45 [0.37, 0.56]	38 [29, 46]	44 [22, 67]
Fixed grid of starts	0.50 [0.41, 0.62]	33 [17, 54]	44 [11, 78]
Whole image	0.56 [0.44, 0.71]	75 [58, 92]	61 [50, 67]

Table 2. Ablations on CAD6 (frozen settings, fixed 48-case subset).

Case	n	Dent claimed	of which located	Declined ("not supported")
Healthy	30	1	–	29
Stain only	60	8	–	52
Visible dent	37	20	20	17
Hidden dent	113	17	11	96

Table 3. Every sealed-test decision of the frozen model ($\tau = 1.52$). A false alarm is a claim on a healthy or stain-only case; an abstention is a decline on a dent: correct when the dent is hidden, a miss when it is visible. The two never overlap.

Operating point	α	τ	Located %	False alarm %	Abstain (hidden) %	Use
Conservative	0.02	1.71	54	10	87	safety-critical sign-off
Balanced (paper)	0.05	1.52	54	10	85	routine inspection
Screening	0.20	0.45	68	22	61	triage, human review

Table 4. Three deployable operating points of the same frozen model. Each threshold is the split-conformal quantile of fit gains on the calibration split at level α; numbers are on the sealed test. With 60 calibration negatives, $\alpha = 0.02$ is the strictest certifiable level.

1. Additional tables

2. Pre-registered negative result (v2)

Targeting the remaining false alarms, we pre-registered three changes and an adoption rule on development blades: a smooth surface appearance field shared by both hypotheses (A), a dent-size prior (D), or both (E), adopted only if recall at a matched false-alarm rate improved by at least five points. None did: D matched v1 (67% recall, AUROC 0.87 vs. 0.85), while A and E raised the fit gains of negatives so much that recall fell to 6%; the extra appearance freedom helped the dent hypothesis as much as the null. The change that does reduce false alarms is the v1.1 reference correction of silhouette and shading, which also lowers the worst development false-alarm gain on Rotor37 from 2.46 to 0.09.

3. Operating points

At a calibration level of $\alpha = 0.2$ (screening), the frozen model locates 68% of visible dents at 22% false alarms.